

THE SYMMETRIC CLOSURE OF THE MAXIMAL NONSYMMETRIC THICK CUBE CATEGORY IS NOT MAXIMAL

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ABSTRACT. In the notation of *Towards a theory of natural directed paths*, let $\widehat{\square}$ be the maximal thick category of cubes which contains no nontrivial coordinate permutation, and let $\widehat{\square}_S$ be the category of all cotransverse maps. We prove that adjoining all coordinate permutations to $\widehat{\square}$ does not generate $\widehat{\square}_S$. An explicit witness is a cotransverse endomorphism of the 4-cube defined from the path graph P_4 . The proof reduces a hypothetical factorization of this map to an ordered edge-colouring of the complete graph on the four vertices of P_4 , compatible with its edge–nonedge partition; a direct four-vertex argument then yields a contradiction. The Lean formalization is available here: <https://github.com/pgs0042/Cotransverse-Maps>.

CONTENTS

1. Cube-theoretic preliminaries	1
2. The graph-theoretic input	2
3. The explicit cotransverse map	4
References	6

1. CUBE-THEORETIC PRELIMINARIES

We use the terminology and notation of [2]. Thus

$$[n] = \{0 < 1\}^n,$$

the box category $\square$ is generated by the coface maps δ_i^α , and a strictly increasing map $f: [m] \rightarrow [n]$ is *cotransverse* when it takes each directed edge to a directed edge. The category of all cotransverse maps is denoted by $\widehat{\square}_S$.

The category $\widehat{\square}$ is characterized in [2, Proposition 2.13] by

$$(1) \quad \widehat{\square} = \left\{ \phi \in \widehat{\square}_S \mid \begin{array}{l} \text{for every } \delta: [p] \rightarrow [m] \text{ in } \square, \\ \phi\delta \text{ one-to-one implies } \phi\delta \in \square \end{array} \right\},$$

where $[m]$ is the domain of ϕ . It is the greatest thick category of cubes containing no nontrivial coordinate permutation [2, Theorem 2.17].

Let $\mathcal{B}$ denote the subcategory of $\widehat{\square}_S$ generated by the maps of $\widehat{\square}$ and by all coordinate permutations. Our goal is to prove

$$\mathcal{B} \subsetneq \widehat{\square}_S.$$

We identify the vertices of $[n]$ with the subsets of $\{1, \dots, n\}$. The vertex corresponding to A is denoted by χ_A and has rank $|A|$.

This work is AI-assisted.

1.1. Lemma (Rank preservation). *Every cotransverse endomorphism $u: [n] \rightarrow [n]$ preserves rank. In particular, it fixes $\chi_\emptyset$ and $\chi_{\{1, \dots, n\}}$.*

Proof. The image of a maximal chain of $[n]$ is a directed path of length n . Since every directed edge raises rank by one, its initial vertex has rank zero and its final vertex has rank n . Every vertex of rank r occurs as the r -th vertex of a maximal chain, and therefore its image also has rank r . $\square$

1.2. Lemma. *Let $u: [n] \rightarrow [n]$ be cotransverse, and define $a_u: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ by*

$$u(\chi_{\{i\}}) = \chi_{\{a_u(i)\}}.$$

The map u is bijective if and only if a_u is bijective. In that case u is the coordinate permutation induced by a_u .

Proof. Only the converse requires proof. Suppose that a_u is bijective. If $i \in A$, strict monotonicity gives

$$\chi_{\{a_u(i)\}} = u(\chi_{\{i\}}) \leq u(\chi_A).$$

Consequently $a_u(A) \subseteq u(\chi_A)$. Both subsets have cardinality $|A|$ by Lemma 1.1; hence $u(\chi_A) = \chi_{a_u(A)}$ for every A . $\square$

By (1), the only bijective element of $\widehat{\square}([n], [n])$ is the identity. This is also stated explicitly in [2, Proposition 2.13].

2. THE GRAPH-THEORETIC INPUT

We recall the graph terminology used in the proof and distinguish the definition of a threshold graph from the characterization that will be applied below. All graphs in this section are finite, simple, and undirected.

A *module* of a graph G is a subset $M \subseteq V(G)$ such that every vertex outside M is adjacent either to every vertex of M or to none of them. The singletons and $V(G)$ are the nonempty trivial modules; we also regard the empty set as a trivial module. A graph whose modules are all trivial is called *prime*. These are the standard notions of modular decomposition; see [3, Section 3.1, p. 24:10] and [1, Definition 2, p. 57]. The convention concerning the empty module does not affect the notion of a prime graph.

The *complement* $\overline{G}$ has vertex set $V(G)$, with two distinct vertices adjacent in $\overline{G}$ precisely when they are not adjacent in G .

The complete graph on a vertex set V has as its edges all two-element subsets of V . An *edge-colouring* of this graph assigns a colour to each such subset. It has a *rainbow triangle* if the three edges of some triangle have three pairwise distinct colours.

For a graph G and $v \in V(G)$, let $N_G(v)$ be the set of vertices adjacent to v , and let $N_G[v] = N_G(v) \cup \{v\}$. The *vicinal preorder* used here is

$$(2) \quad v \preceq_G w \iff N_G(v) \subseteq N_G[w].$$

It is *total* when any two vertices are comparable: for all $v, w \in V(G)$, either $v \preceq_G w$ or $w \preceq_G v$.

2.1. Definition (Threshold graph). A subset $U \subseteq V(G)$ is *independent* if no two of its vertices are adjacent. A graph G is a *threshold graph* if there exist a weight function

$w: V(G) \rightarrow \mathbb{R}$ and a threshold $t \in \mathbb{R}$ such that, for every $U \subseteq V(G)$,

$$U \text{ is independent} \iff \sum_{v \in U} w(v) \leq t.$$

This is the definition in [4, Definition 2].

The preceding weight condition is the definition. The proof below uses the following classical characterization.

2.2. Theorem (Vicinal characterization of threshold graphs). *A finite graph is a threshold graph if and only if its vicinal preorder is total [4, Section 3].*

Klivans uses the opposite orientation for the vicinal preorder in [4, Definition 3]. Reversing a preorder does not affect its totality, so Theorem 2.2 has the form stated above with convention (2).

Let P_4 be the path graph with vertex set $V = \{1, 2, 3, 4\}$ and edge set

$$(3) \quad E(P_4) = \{\{1, 2\}, \{2, 3\}, \{3, 4\}\}.$$

2.3. Lemma (Four-vertex colouring obstruction). *Let λ be an edge-colouring of the complete graph on $V = \{1, 2, 3, 4\}$. Suppose that every colour class consists entirely of edges of P_4 or entirely of nonedges of P_4 . If λ uses exactly three colours, then it has a rainbow triangle.*

Proof. Assume for a contradiction that there is no rainbow triangle. The sets of colours used on edges and on nonedges of P_4 are disjoint and nonempty. Since their cardinalities add up to three, one of these sets has one element and the other has two.

Suppose first that the three edges $12, 23, 34$ of P_4 have the same colour α , while the nonedges use two colours, neither equal to α . The triangle on $\{1, 2, 4\}$ is not rainbow only if $\lambda(14) = \lambda(24)$, and the triangle on $\{1, 3, 4\}$ is not rainbow only if $\lambda(13) = \lambda(14)$. Thus all three nonedges $13, 14, 24$ have the same colour, a contradiction.

Suppose instead that the three nonedges have the same colour α , while the edges of P_4 use two colours, neither equal to α . The triangle on $\{1, 2, 3\}$ is not rainbow only if $\lambda(12) = \lambda(23)$, and the triangle on $\{2, 3, 4\}$ is not rainbow only if $\lambda(23) = \lambda(34)$. Hence all three edges of P_4 have the same colour, again a contradiction. $\square$

2.4. Lemma. *The graph P_4 is prime, is isomorphic to its complement, and is not a threshold graph.*

Proof. Writing ij for $\{i, j\}$, a vertex distinguishing the two members of each two-element subset can be chosen as follows:

M	12	13	14	23	24	34
distinguishing vertex	3	4	2	1	1	2.

Thus no two-element subset is a module. If M has three elements, the remaining vertex has degree one or two in P_4 , so it is adjacent to some but not all vertices of M . Hence M is not a module either, and P_4 is prime. Its complement is the path $3 - 1 - 4 - 2$.

Finally, vertices 1 and 4 are incomparable in the vicinal preorder:

$$N_{P_4}(1) = \{2\} \not\supseteq \{3, 4\} = N_{P_4}[4], \quad N_{P_4}(4) = \{3\} \not\supseteq \{1, 2\} = N_{P_4}[1].$$

The vicinal preorder of P_4 is therefore not total, and Theorem 2.2 shows that P_4 is not a threshold graph. $\square$

3. THE EXPLICIT COTRANSVERSE MAP

Define $F: [4] \rightarrow [4]$ by the following unambiguous formula:

$$(4) \quad F(\chi_A) = \begin{cases} \chi_\emptyset, & |A| = 0, \\ \chi_{\{1\}}, & |A| = 1, \\ \chi_{\{1,2\}}, & |A| = 2 \text{ and } A \in E(P_4), \\ \chi_{\{1,3\}}, & |A| = 2 \text{ and } A \notin E(P_4), \\ \chi_{\{1,2,3\}}, & |A| = 3, \\ \chi_{\{1,2,3,4\}}, & |A| = 4. \end{cases}$$

3.1. Proposition. *The map F belongs to $\widehat{\square}_S([4], [4])$.*

Proof. For every cover $A \subset A \cup \{i\}$, the corresponding values in (4) form one of the inclusions

$$\emptyset \subset \{1\} \subset \{1, 2\} \text{ or } \{1, 3\} \subset \{1, 2, 3\} \subset \{1, 2, 3, 4\}.$$

Each inclusion adds exactly one coordinate. Thus F is strictly increasing and takes directed edges to directed edges. $\square$

3.2. Theorem. *The map F is not a composition of maps of $\widehat{\square}$ and coordinate permutations. Consequently,*

$$\mathcal{B}([4], [4]) \subsetneq \widehat{\square}_S([4], [4]) \quad \text{and} \quad \mathcal{B} \subsetneq \widehat{\square}_S.$$

Proof. Suppose, for a contradiction, that $F \in \mathcal{B}$. There are no cotransverse maps $[m] \rightarrow [n]$ for $m > n$: the image of a maximal chain would otherwise be a directed path of length m in $[n]$. The dimensions in a decomposition of the endomorphism $F: [4] \rightarrow [4]$ therefore form a nondecreasing sequence which starts and ends at 4. Hence every intermediate object is $[4]$.

Read a decomposition in the order in which its factors are applied, and let $h \in \widehat{\square}([4], [4])$ be the first non-bijective factor. Such a factor exists because (4) sends all four rank-one vertices to the same vertex. All preceding factors from $\widehat{\square}$ are identities, and all other preceding factors are coordinate permutations. Their composite is therefore a coordinate permutation s . Thus we can write

$$(5) \quad F = ghs = gu,$$

where $u = hs$ and g is the composite of the remaining factors. In particular,

$$(6) \quad u(x) = u(y) \implies F(x) = F(y).$$

For clarity, let a_u be the rank-one map of Lemma 1.2. Since u is non-bijective, a_u has a fibre C with at least two elements. We claim that C is a module of P_4 . Indeed, take $p, q \in C$ and $k \notin C$. The two singleton images indexed by $a_u(p) = a_u(q)$ and $a_u(k)$ are distinct. Rank preservation and monotonicity force

$$u(\chi_{\{p,k\}}) = \chi_{\{a_u(p), a_u(k)\}} = u(\chi_{\{q,k\}}).$$

By (6), the pairs $\{p, k\}$ and $\{q, k\}$ have the same image under F . Formula (4) says exactly that k is adjacent to p in P_4 if and only if it is adjacent to q . Thus C is a module. Lemma 2.4 implies $C = V(P_4) = V = \{1, 2, 3, 4\}$, so there is an $a \in \{1, 2, 3, 4\}$ such that

$$(7) \quad u(\chi_{\{i\}}) = \chi_{\{a\}} \quad (1 \leq i \leq 4).$$

For every unordered pair $ij = \{i, j\}$, rank preservation now gives a unique colour

$$\kappa(ij) \in D := \{1, 2, 3, 4\} \setminus \{a\} \quad \text{such that} \quad u(\chi_{ij}) = \chi_{\{a, \kappa(ij)\}}.$$

Regard κ as an edge-colouring of the complete graph on V . If two pairs have the same κ -colour, they have the same image under u and hence, by (6), the same image under F . Therefore every κ -colour class consists entirely of edges of P_4 or entirely of nonedges of P_4 .

The colouring κ has no rainbow triangle. Indeed, for three distinct vertices i, j, k , the rank-three set $u(\chi_{\{i, j, k\}})$ contains each of

$$\{a, \kappa(ij)\}, \quad \{a, \kappa(ik)\}, \quad \{a, \kappa(jk)\}.$$

If the three colours were distinct, their union would have four elements, whereas $u(\chi_{\{i, j, k\}})$ has rank three. Thus there is no rainbow triangle.

We also record the order constraint supplied by $h \in \hat{\square}$. Let π be the permutation of coordinate indices induced by s , and order V by

$$i \prec j \iff \pi(i) < \pi(j).$$

The set D retains the usual order of coordinate indices. If $i \prec j \prec k$, then

$$(8) \quad \kappa(ij) \leq \kappa(ik) \leq \kappa(jk).$$

To prove the first inequality, consider the 2-face of [4] whose variable coordinates are $\pi(j) < \pi(k)$, whose coordinate $\pi(i)$ is fixed at 1, and whose remaining coordinate is fixed at 0. If $\kappa(ij) \neq \kappa(ik)$, the restriction of h to this face is one-to-one: its two middle vertices have distinct images and the other two images have different ranks. By (1), this restriction belongs to $\square$. A coface map preserves the order of its variable coordinates, so $\kappa(ij) < \kappa(ik)$. Equality gives the weak inequality in the remaining case. The second inequality follows in the same way from the face whose variable coordinates are $\pi(i) < \pi(j)$ and whose coordinate $\pi(k)$ is fixed at 1.

The set D has three elements. The colouring cannot use only one colour, because it refines the nonconstant edge–nonedge colouring of P_4 . If it used all three colours, the four-vertex colouring obstruction, Lemma 2.3, would produce a rainbow triangle, contrary to the preceding paragraph.

It remains to suppose that κ uses exactly two colours. One of them colours precisely the edges of P_4 and the other precisely its nonedges. Let H be the graph whose edges are the pairs bearing the larger colour. Thus H is either P_4 or its complement, and hence $H \cong P_4$. We show that the vicinal preorder of H is total. If $i \prec j$ and $k \in N_H(i) \setminus \{j\}$, apply (8) to the three vertices in their $\prec$ -order. In each of the three possible positions of k , the colour of jk is greater than or equal to the colour of ik . Therefore jk is also an edge of H , and

$$N_H(i) \subseteq N_H[j].$$

Thus $i \preceq_H j$ whenever $i \prec j$, so the vicinal preorder is total. Theorem 2.2 implies that H is a threshold graph. Transporting the vertex weights in Definition 2.1 shows that this property is invariant under graph isomorphism. Since $H \cong P_4$, this contradicts Lemma 2.4.

All possible numbers of colours lead to a contradiction. Therefore $F \notin \mathcal{B}$, and Proposition 3.1 completes the proof. $\square$

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